Programming in C Lab

Record Programs

Program 1 21/01/2026

Aim: To write a program to demonstrate the Grading of students using the if, if-else, if-else-if-else and nested-if constructs

Source Code:

```
#include <stdio.h>

int main ()

{

   int marks;

   char category;

   printf("Enter student marks (0 - 100): ");

   scanf("%d", &marks);

   printf("Enter category (G for General, R for Reserved): ");

   scanf(" %c", &category);

   /* --------- IF - ELSE IF --------- */

   if (marks >= 90)

      printf("\nGrade: A+");

   else if (marks >= 75)

      printf("\nGrade: A");

   else if (marks >= 60)

      printf("\nGrade: B");

   else if (marks >= 40)

      printf("\nGrade: C");

   else

      printf("\nGrade: Fail");

   /* --------- NESTED IF --------- */
```

```
    if (marks >= 60)
    {
        if (category == 'R' || category == 'r')
            printf ("\nScholarship Status: Eligible");
        else
            printf ("\nScholarship Status: Not Eligible");
    }
    else
    {
        printf ("\nScholarship Status: Not Eligible");
    }
    return 0;
}
```

Output: Should be written on the left side paper of the rough as well as fair record

Result: Program is executed successfully

Program 2 21/01/2026

Aim: To write a program to demonstrate the use of switch-case construct to display the name of month

Source Code:

```
#include <stdio.h>

int main()

{

   int x;

   printf("Enter a number:\n");

   printf("The Month Name Is:\n");

   scanf("%d", &x);

   switch (x)

   {

      case 1:  printf("January"); break;

      case 2:  printf("February"); break;

      case 3:  printf("March"); break;

      case 4:  printf("April"); break;

      case 5:  printf("May"); break;

      case 6:  printf("June"); break;

      case 7:  printf("July"); break;

      case 8:  printf("August"); break;

      case 9:  printf("September"); break;

      case 10: printf("October"); break;

      case 11: printf("November"); break;

      case 12: printf("December"); break;

      default: printf("Enter a Valid Number (1-12)");

   }
```

```
    return 0;
}
```

Output:

Result:

Program 3 **Date:**

Aim: To write a program to evaluate whether a number is a twisted prime number with the use of while loop

Source code:

```
#include <stdio.h>

int main()

{

   int n, reversed = 0, remainder, i, flag = 0;

   // Input number

   printf("Enter a number: ");

   scanf("%d", &n);

   // Check if n is prime

   if (n <= 1)

   {

      printf("%d is not a Twisted Prime Number.\n", n);

      return 0;

   }

   i = 2;

   flag = 0;

   while (i <= n / 2)

   {

      if (n % i == 0)

      {

         flag = 1;

         break;

      }

      i++;
```

```
}
if (flag == 1)
{
    printf("%d is not a Twisted Prime Number because it is not prime.\n", n);
    return 0;
}
// Reverse the number
int temp = n;
while (temp!= 0)
{
    remainder = temp % 10;
    reversed = reversed * 10 + remainder;
    temp /= 10;
}
// Check if reversed number is prime
i = 2;
flag = 0;
while (i <= reversed / 2)
{
    if (reversed % i == 0)
    {
        flag = 1;
        break;
    }
    i++;
}
if (flag == 0)
```

```
        printf("%d is a Twisted Prime Number.\n", n);
    else
        printf("%d is not a Twisted Prime Number because its reverse (%d) is not prime.\n", n,
reversed);
    return 0;
}
```

Output:

Result: The program is successfully executed

Program 4 **Date:**

Aim: To write a program to demonstrate the use of do-while loop using a menu driven program

Source Code:

```
#include <stdio.h>
int main()
{
   int choice;
   double num1, num2, result;
   do {
      printf("\n==== MENU-DRIVEN CALCULATOR ====\n");
      printf("1. Addition\n");
      printf("2. Subtraction\n");
      printf("3. Multiplication\n");
      printf("4. Division\n");
      printf("5. Exit\n");
      printf("Enter your choice (1-5): ");
      scanf("%d", &choice);
      if (choice >= 1 && choice <= 4)
      {
         printf("Enter two numbers: ");
         scanf("%lf %lf", &num1, &num2);
      }
      switch (choice)
      {
         case 1:
            result = num1 + num2;
            printf("Result: %.2lf + %.2lf = %.2lf\n", num1, num2, result);
```

```
            break;
        case 2:
            result = num1 - num2;
            printf("Result: %.2lf - %.2lf = %.2lf\n", num1, num2, result);
            break;
        case 3:
            result = num1 * num2;
            printf("Result: %.2lf * %.2lf = %.2lf\n", num1, num2, result);
            break;
        case 4:
            if (num2 == 0)
                printf("Error! Division by zero is not allowed.\n");
            else
            {
                result = num1 / num2;
                printf("Result: %.2lf / %.2lf = %.2lf\n", num1, num2, result);
            }
            break;
        case 5:
            printf("Exiting program... Goodbye!\n");
            break;
        default:
            printf("Invalid choice! Please select 1–5.\n");
    }
  } while (choice != 5);
  return 0;
}
```

Output:

Result:

Program 5 **Date:**

Aim: To write a program to demonstrate the use of for loop to find whether a given number is:

a. Odd or even
b. Prime or not and
c. Compute the sum of all numbers

Source Code:

```
#include <stdio.h>

#include <math.h>

int main() {

   int n, i, j, isPrime, sum = 0;

   printf("Enter a number: ");

   scanf("%d", &n);

   // Single complex for loop

   for (i = 1; i <= n; i++) {

      sum += i;  // accumulating sum

      // Check even or odd

      if (i % 2 == 0)

         printf("%d is Even", i);

      else

         printf("%d is Odd", i);

      // Check if i is prime (using inner logic, not another for)

      isPrime = 1;  // assume prime

      if (i <= 1)

         isPrime = 0;

      else {

         // Instead of another for loop, use arithmetic trick

         // Check divisibility by 2, 3, 5, 7 — limited primality test
```

```
        if ((i % 2 == 0 && i != 2) || (i % 3 == 0 && i != 3) ||
            (i % 5 == 0 && i != 5) || (i % 7 == 0 && i != 7))
            isPrime = 0;
    }
    if (isPrime)
        printf(" and Prime");
    else
        printf(" and Not Prime");
    printf("\n");
  }
  printf("\nSum of numbers from 1 to %d = %d\n", n, sum);
  return 0;
}
```

Output:

Result:

Program 6 **Date:**

Aim: To write a program to demonstrate the use of break and continue using for loop

Source Code:

```
#include <stdio.h>
int main() {
    int n, i, sum = 0;
    printf("Enter the limit (n): ");
    scanf("%d", &n);
    printf("\nProcessing numbers from 1 to %d...\n\n", n);
    for (i = 1; i <= n; i++) {
        // If number is multiple of 5, skip it
        if (i % 5 == 0) {
            printf("Skipping %d (multiple of 5)\n", i);
            continue;  // skips rest of the loop body
        }
        // If number exceeds 50, break the loop completely
        if (i > 50) {
            printf("\nNumber exceeded 50 → breaking the loop!\n");
            break;  // exits the loop
        }
        // Add only even numbers to the sum
        if (i % 2 == 0) {
            sum += i;
            printf("%d is even → added to sum (current sum = %d)\n", i, sum);
        } else {
            printf("%d is odd → ignored for sum\n", i);
        }
```

```
    }
    printf("\nFinal Sum of Even Numbers (excluding multiples of 5 and numbers > 50): %d\n",
sum);
    return 0;
}
```

Output:

Result:

Program 7 **Date:**

Aim: To write a program to demonstrate the use of nested loops for performing the sum rows and columns

Source Code:

```
#include <stdio.h>

int main() {

    int rows, cols;

    printf("Enter number of rows: ");

    scanf("%d", &rows);

    printf("Enter number of columns: ");

    scanf("%d", &cols);

    printf("\n--- Multiplication Table (%d x %d) ---\n\n", rows, cols);

    int i, j, product, rowSum, colSum;

    // Print header row

    printf("     ");

    for (j = 1; j <= cols; j++)

        printf("%4d", j);

    printf("\n---------------------------------------------\n");

    // Outer loop for rows

    for (i = 1; i <= rows; i++) {

        rowSum = 0;

        printf("%2d | ", i);

        // Inner loop for columns

        for (j = 1; j <= cols; j++) {

            product = i * j;

            rowSum += product;

            // Display different symbols for even/odd
```

```
            if (product % 2 == 0)
                printf("%4d", product);  // even product
            else
                printf("%3d*", product); // odd product (marked with *)
        }
        printf("   | Row Sum = %d\n", rowSum);
    }
    // Calculate and print column sums
    printf("--------------------------------------------\n");
    printf("Sum:");
    for (j = 1; j <= cols; j++) {
        colSum = 0;
        for (i = 1; i <= rows; i++) {
            colSum += i * j;
        }
        printf("%4d", colSum);
    }
    printf("\n");
    return 0;
}
```

Output:

Result:

Program 8 **Date:**

Aim: To write a program to demonstrate the use of strings. The program will perform:

a. Declare multiple strings
b. Read a full sentence (including spaces)
c. Display the following values:
    a. Original String
    b. Length of string
    c. String in uppercase
    d. String reversed

Source Code:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int main() {

    char str[100], upper[100], reverse[100];

    int i, len;

    // Read string input (including spaces)

    printf("Enter a sentence: ");

    fgets(str, sizeof(str), stdin);

    // Remove newline character if present

    str[strcspn(str, "\n")] = '\0';

    printf("\n--- String Details ---\n");

    // 1 Display the original string

    printf("Original String: %s\n", str);

    // 2 Calculate and display length

    len = strlen(str);

    printf("Length of String: %d\n", len);

    // 3 Convert to uppercase
```

```
    for (i = 0; i < len; i++) {
        upper[i] = toupper(str[i]);
    }
    upper[i] = '\0';  // end of string
    printf("Uppercase: %s\n", upper);
    // 4 Reverse the string
    for (i = 0; i < len; i++) {
        reverse[i] = str[len - i - 1];
    }
    reverse[i] = '\0';
    printf("Reversed: %s\n", reverse);
    // 5 Count vowels
    int vowels = 0;
    for (i = 0; i < len; i++) {
        char ch = tolower(str[i]);
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
            vowels++;
    }
    printf("Number of vowels: %d\n", vowels);
    return 0;
}
```

Output:

Result:

Program 9 **Date:**

Aim: To write a program to demonstrate the use of arrays. The program will perform the following operations:

- Compute the **sum** and **average**
- Find **maximum** and **minimum**
- Sort the array in **ascending order**
- **Search** for an element using **linear search**

Source Code:

```
#include <stdio.h>

int main() {

    int arr[100], n, i, j, temp, sum = 0, search, found = 0;

    printf("Enter number of elements (max 100): ");

    scanf("%d", &n);

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++) {

        scanf("%d", &arr[i]);

        sum += arr[i];  // sum while reading

    }

    printf("\n--- Array Operations ---\n");

    // 1 Display original array

    printf("Original Array: ");

    for (i = 0; i < n; i++) {

        printf("%d ", arr[i]);

    }

    // 2 Calculate average

    float avg = (float)sum / n;

    printf("\nSum = %d, Average = %.2f\n", sum, avg);
```

```
// 3 Find max and min
int max = arr[0], min = arr[0];
for (i = 1; i < n; i++) {
    if (arr[i] > max)
        max = arr[i];
    if (arr[i] < min)
        min = arr[i];
}
printf("Maximum = %d, Minimum = %d\n", max, min);
// 4 Sort the array (Bubble Sort)
for (i = 0; i < n - 1; i++) {
    for (j = 0; j < n - i - 1; j++) {
        if (arr[j] > arr[j + 1]) {
            temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
        }
    }
}
printf("Sorted Array (Ascending): ");
for (i = 0; i < n; i++) {
    printf("%d ", arr[i]);
}
// 5 Search for an element
printf("\nEnter an element to search: ");
scanf("%d", &search);
for (i = 0; i < n; i++) {
```

```
        if (arr[i] == search) {
            printf("Element %d found at position %d\n", search, i + 1);
            found = 1;
            break;
        }
    }
    if (!found)
        printf("Element %d not found in the array.\n", search);
    printf("\n");
    return 0;
}
```

Output:

Result:

Program 10 **Date:**

Aim: To write a program to demonstrate the addition of two arrays.

Source Code:

```
#include <stdio.h>
int main() {
    int A[10][10], B[10][10], Sum[10][10];
    int rows, cols, i, j;
    // Input dimensions
    printf("Enter number of rows (max 10): ");
    scanf("%d", &rows);
    printf("Enter number of columns (max 10): ");
    scanf("%d", &cols);
    // Input elements of first matrix
    printf("\nEnter elements of Matrix A:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            printf("A[%d][%d] = ", i + 1, j + 1);
            scanf("%d", &A[i][j]);
        }
    }
    // Input elements of second matrix
    printf("\nEnter elements of Matrix B:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            printf("B[%d][%d] = ", i + 1, j + 1);
            scanf("%d", &B[i][j]);
        }
```

```
    }
    // Matrix addition
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            Sum[i][j] = A[i][j] + B[i][j];
        }
    }
    // Display result
    printf("\n--- Resultant Matrix (A + B) ---\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            printf("%5d", Sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

Result:

Program 11 **Date:**

Aim: To write a program to demonstrate the multiplication of two arrays.

Source Code:

```
#include <stdio.h>
int main() {
    int A[10][10], B[10][10], C[10][10];
    int i, j, k, r1, c1, r2, c2;
    // Input dimensions of matrices
    printf("Enter rows and columns for Matrix A: ");
    scanf("%d %d", &r1, &c1);
    printf("Enter rows and columns for Matrix B: ");
    scanf("%d %d", &r2, &c2);
    // Check if multiplication is possible
    if (c1 != r2) {
        printf("\nMatrix multiplication not possible!\n");
        printf("Columns of A must be equal to rows of B.\n");
        return 0;
    }
    // Input elements of Matrix A
    printf("\nEnter elements of Matrix A:\n");
    for (i = 0; i < r1; i++) {
        for (j = 0; j < c1; j++) {
            printf("A[%d][%d] = ", i + 1, j + 1);
            scanf("%d", &A[i][j]);
        }
    }
    // Input elements of Matrix B
```

```
printf("\nEnter elements of Matrix B:\n");
for (i = 0; i < r2; i++) {
    for (j = 0; j < c2; j++) {
        printf("B[%d][%d] = ", i + 1, j + 1);
        scanf("%d", &B[i][j]);
    }
}
// Initialize result matrix to 0
for (i = 0; i < r1; i++) {
    for (j = 0; j < c2; j++) {
        C[i][j] = 0;
    }
}
// Perform matrix multiplication
for (i = 0; i < r1; i++) {
    for (j = 0; j < c2; j++) {
        for (k = 0; k < c1; k++) {
            C[i][j] += A[i][k] * B[k][j];
        }
    }
}
// Display result
printf("\n--- Resultant Matrix (A × B) ---\n");
for (i = 0; i < r1; i++) {
    for (j = 0; j < c2; j++) {
        printf("%5d", C[i][j]);
    }
```

```
        printf("\n");
    }
    return 0;
}
```

Output:

Result:

Program 12 **Date:**

Aim: To write a program to demonstrate the